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Modern Communication Systems F25

Lab 7

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Theory:

1) Angle of Arrival (AoA) True/False

a) True

A single two antenna pair gives a bearing only, you need to find the intersections of bearings from two spatially separated receivers for localization

b) False

Even with no noise/multipath, large antenna spacing causes phase warping/spatial aliasing. The measured phase only gives $s \cos \theta / \lambda$ modulo an integer, so the direction is ambiguous unless you limit s or use multiple spacings

c) True

Strong multipath adds components with different delays/phases, and the phase difference can be dominated by these reflections, so the simple planar wave relation becomes unreliable unless you separate paths or somehow mitigate multipath

d) False

A transmitter-receiver CFO produces a common time varying phase rotation applied equally to both antennas of the same receiver, so the relative phase is unchanged. CFO cancels out instantaneously. You do need to handle CFO if you measure phases at different times or if receive chains are not mutually synchronized

2) Angle-of-Arrival (AoA) Estimation and Array Ambiguities

a) Phase difference model

for a ULA with two elements separated by s and a plane wave arriving at angle θ , the extra path traveled to the second element relative to the first is $d = s \cos \theta$. A narrowband carrier converts that extra distance into a phase delay $\Delta \Phi = 2\pi d / \lambda = 2\pi s \cos(\theta) / \lambda$. Additionally, phases are inherently mod 2π .

This assumes:

far field plane wave propagation so $d = s \cos \theta$ holds
narrowband signal so a single carrier phase maps distance linearly
single dominant path so we don't get multipath or reflection issues
identical synchronized receiver (no unknown antenna phase offsets)
negligible mutual coupling and identical antenna patterns

b) Ambiguity condition

if $s > \lambda/2$, the phase can span $2\pi s / \lambda * (\max \cos - \min \cos) = 2\pi s / \lambda * 2 = 4\pi$ ($s > \lambda/2$) / $\lambda = 2\pi (>1) > 2\pi$, and since phase is mod 2π , this range greater than 2π overlaps itself, resulting in spatial aliasing.

Choosing $s = \lambda/2$ gives $\Delta \Phi$ a range of $[-\pi, \pi]$, the max range without overlap

c) Numerical check

$f = 5 \times 10^9$, $\lambda = 3 \times 10^8 / 5 \times 10^9 = 0.06\text{m}$

$\cos(30) = \sqrt{3}/2 \approx .8660$

$s = \lambda/2$

$\Delta \Phi = 2\pi s / \lambda \cos \theta = 2\pi \cdot 1/2 * .8660 = \pi * .8660 = 2.720699 \text{ rad}$,

within $[-\pi, \pi]$, no wrapping

$s = \lambda = 0.06\text{m}$

$\Delta \Phi = 2\pi \cdot 1 * .866 = 5.44 \text{ rad}$, greater than π ,

subtract 2π for $-.84178 \text{ rad}$

attempts to invert using $\cos(\theta) = -.8417/2\pi = -.1339$

$\theta = \arccos(-.1339) = 97.69 \text{ degrees}$

with $s = \lambda$ the same measured wrapped phase corresponds to both 30 and 97.7 degrees, could be either

3) ArrayTrack Multipath Separation with AoA and TOF

a) Why strongest path is not always LoS

1. constructive reflection/focusing. walls can direct energy toward the receiver and make a stronger path than the direct LoS path from a different angle
2. Obstruction/shadowing of LoS. Furniture, walls, or people can block the LoS and make that path relatively quiet compared to other reflected paths.

b) Combining AoA and TOF to separate paths

i) the AoA spectrum

Compute the spatial spectrum from the antenna array

Identify multiple peaks corresponding to distinct incoming directions

ii) the multipath power-delay profile, and

For each AoA peak, estimate the TOF by fitting the phase slope across OFDM subcarriers
this gives delay of each spatial path, giving AoA vs delay

iii) relative delays between paths,

select LoS as the earliest arriving path.

if multiple arrive from same angle, choose one with minimum delay

use both AoA and delay: AoA tells bearing and delay tells distance. Peaks early in time that are consistent with expected direction are the LoS

c) Delay resolution

$$\Delta \tau_{\min} \approx 1/(20 \times 10^6) \approx 50 \text{ ns}$$

$$c \cdot \Delta \tau_{\min} = 3 \times 10^8 \cdot 50 \times 10^{-9} = 15 \text{ meters}$$

so the temporal resolution is limited by the bandwidth of the system

d) Explain

path length difference = 1m

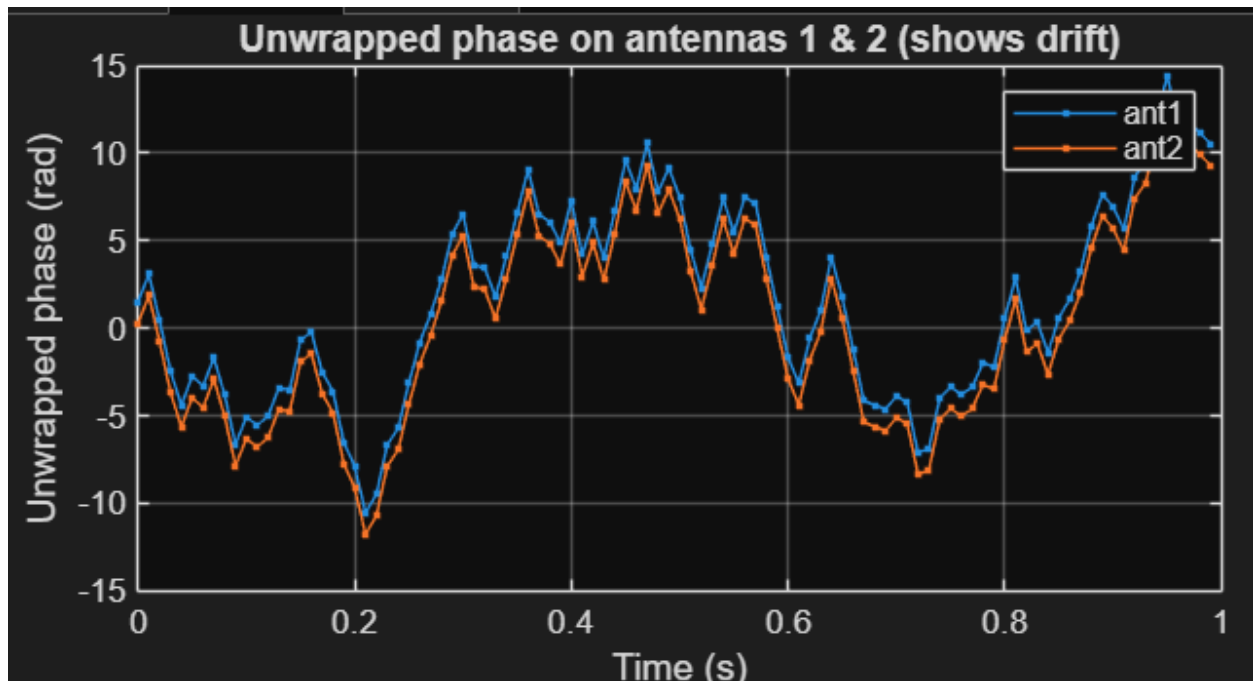
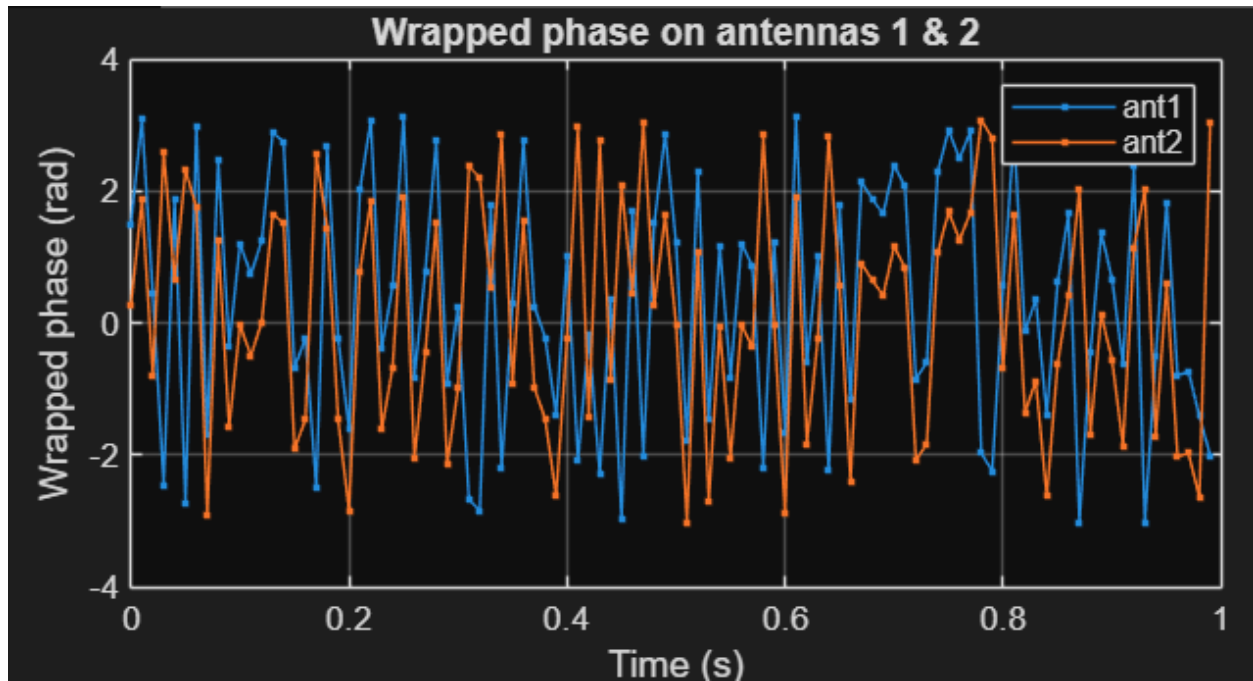
$$\Delta \tau = \Delta d / c = 1/3 \times 10^8 = 3.33 \text{ ns}$$

the 1m path is much smaller than the system's delay resolution, and it can't reliably distinguish these two paths

MATLAB:

4) Wireless Channels and Phase Stability

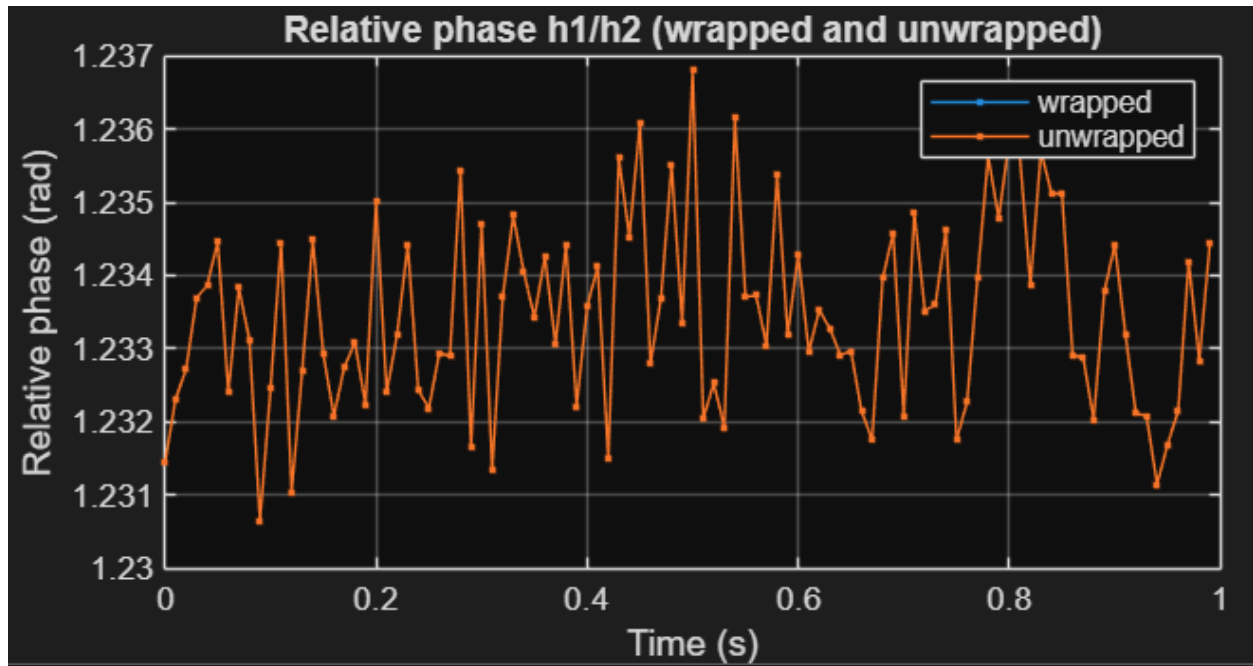
a) Plot the channel phase on antenna 1 and antenna 2



b) Explain the observed phase drift

the drift is caused by CFO, sampling clock mismatch, and phase noise in the RF chain

c) Plot the relative phase between antennas



d) Interpretation of relative phase stability

yes, it's roughly constant over time since both antennas share the same RF hardware and everything is stationary

e) Estimate angle of arrival (AOA) from relative phase

$$\Delta\phi = 2\pi s \cos(\theta) / \lambda$$

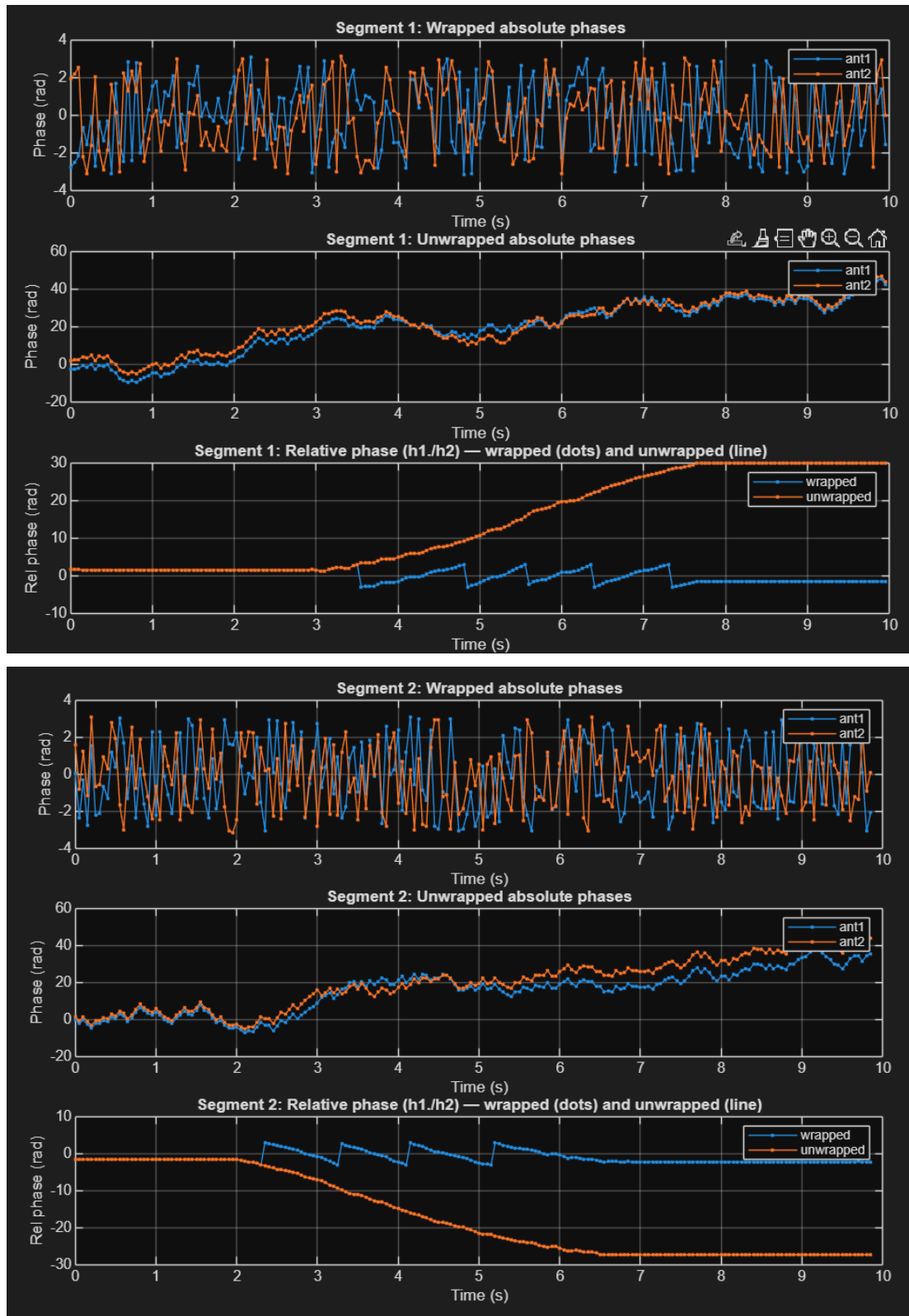
$$\theta = \arccos(\Delta\phi \lambda / 2 \pi s)$$

doing this in matlab code with averages gives:

66.88 degrees

5) Channel Phase and Motion Detection

a) phase over time



b) Direction of motion

Segment 1: right to left <<< towards antenna 1

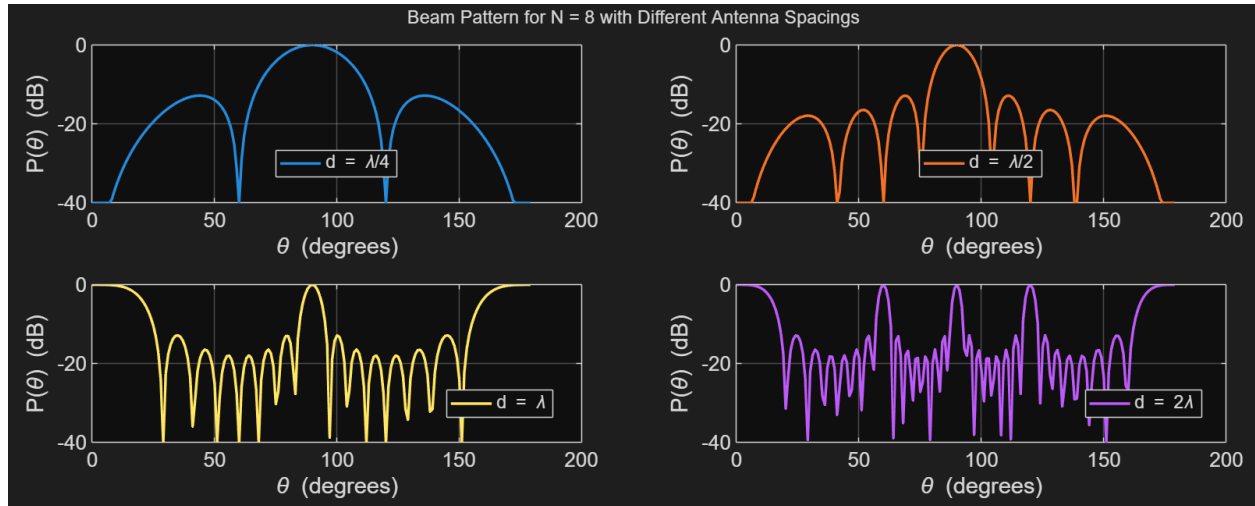
Segment 2: left to right >>> towards antenna 2

c) explanation

relative phase measures how much 1 leads 2. If it is increasing, 1 is leading more relative to 2, must be approaching 1 faster than 2, away from 2 towards 1, right to left

6) Beam Pattern in a ULA

a) MATLAB beam pattern for different spacings



b) Ambiguity and grating lobes

$d = \lambda$ and $d = 2\lambda$ exhibit ambiguity as they have grating lobes, while $d = \lambda/2$ and $d = \lambda/4$ have an obvious single main peak

Increasing spacing past $\lambda/2$ causes this because with $\Delta\phi = 2\pi s \cos(\theta)$, if s is larger than $\lambda/2$, the range of phase differences can exceed 2π , creating an overlap, where two different angles can cause the same phase difference